

Module 17: Reproductive System and Development

Topics: Male Reproductive System, Female Reproductive System, Pregnancy and Development

TOPIC 1 OF 3 . WEEK 20 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Male Reproductive System

Testis structure, spermatogenesis, accessory glands, and the hormonal axis.

The Science

Testes do two jobs: produce sperm and produce testosterone. Sperm production happens in the seminiferous tubules, which are lined with Sertoli cells that support developing spermatogonia through meiosis. Testosterone is produced by Leydig cells (interstitial cells) outside the tubules. The blood-testis barrier (formed by Sertoli cell tight junctions) protects developing sperm from immune attack.

Sperm pathway: seminiferous tubule, rete testis, epididymis (sperm matures here, becomes motile, is stored), vas deferens (ascending through the inguinal canal), ejaculatory duct (joining with seminal vesicle duct), prostatic urethra, urethra. Accessory glands add to the seminal fluid: seminal vesicles (fructose, prostaglandins, about 60% of volume), prostate (alkaline secretion, enzymes, about 30%), bulbourethral glands (mucus pre-ejaculate).

Hormonal axis: hypothalamic GnRH drives pituitary LH and FSH. LH stimulates Leydig cells to make testosterone. FSH plus testosterone supports Sertoli cells and spermatogenesis. Testosterone feeds back negatively on the hypothalamus and pituitary to stabilize the system. Inhibin from Sertoli cells specifically suppresses FSH, fine-tuning spermatogenesis.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cross-section of a testis with seminiferous tubules and Leydig cells labeled, then trace the path of sperm from tubule to urethra.

Common misconceptions to address

- Students think the testis is one structure with one function. It does sperm production AND testosterone production via different cells.
- The epididymis is sometimes overlooked. Sperm leaving the seminiferous tubules are not yet motile; they mature in the epididymis.
- Testosterone is thought of as 'the male hormone' but it is also made (in smaller amounts) by the adrenal cortex in both sexes.

Order of operations (what to teach first)

Testis structure and its two jobs, then sperm pathway with accessory glands, then the hormonal axis.

Self-test prompts

- Which cells produce testosterone?
- Which cells support developing sperm?
- Why do anabolic steroid users develop testicular atrophy?

Why This Matters to Your Patients

Testicular torsion is a surgical emergency; twisting of the spermatic cord cuts off blood supply and the testis dies within hours. Varicocele (enlarged pampiniform plexus) is a common cause of infertility because the heat impairs spermatogenesis. Benign prostatic hyperplasia (BPH) is age-related prostate enlargement causing urinary symptoms. Prostate cancer is the most common solid tumor in men; PSA screening is controversial but widely used. Hypogonadism (from various causes) produces low testosterone, low energy, reduced libido, reduced muscle mass; replacement therapy is increasingly common but has risks.

TOPIC 2 OF 3 . WEEK 20 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Female Reproductive System

Ovary, oogenesis, the ovarian and uterine cycles, and the hormonal choreography.

The Science

Ovaries produce eggs (oocytes) and hormones. Females are born with all their primordial follicles already arrested in prophase I of meiosis. Each cycle a cohort of follicles begins maturing; usually one becomes dominant. The dominant follicle's granulosa and theca cells produce estrogen under FSH and LH stimulation.

Ovarian cycle (about 28 days). Follicular phase (days 1-14): follicle matures; estrogen rises. Ovulation (day 14): LH surge triggers release of the secondary oocyte. Luteal phase (days 15-28): ruptured follicle becomes the corpus luteum, which produces progesterone (and some estrogen) to support potential implantation. If no pregnancy, the corpus luteum regresses, hormones drop, and the cycle restarts.

Uterine (menstrual) cycle parallels the ovarian: menstrual phase (days 1-5, shedding of endometrium), proliferative phase (days 6-14, estrogen rebuilds endometrium), secretory phase (days 15-28, progesterone makes endometrium glandular and receptive). If no implantation, progesterone falls and the cycle restarts.

Hormonal control: GnRH from hypothalamus drives LH and FSH. Estrogen and progesterone feed back, mostly negatively. The key twist: rising estrogen mid-cycle switches feedback from negative to positive, triggering the LH surge. After ovulation, the corpus luteum's progesterone restores negative feedback. If pregnancy occurs, the embryo's hCG keeps the corpus luteum alive until the placenta takes over hormone production.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the ovarian cycle phases with FSH, LH, estrogen, and progesterone curves over 28 days. Then label the uterine phases underneath.

Common misconceptions to address

- Students think eggs are made monthly. All eggs are present from birth; the monthly cycle is which one matures.
- Estrogen is thought to only feed back negatively. The mid-cycle switch to positive feedback is the key trigger for ovulation.
- The menstrual cycle is reduced to the bleeding phase. It is the integration of ovarian and uterine cycles over the entire month.

Order of operations (what to teach first)

Ovarian anatomy and oogenesis, then the ovarian cycle, then the uterine cycle in parallel, then hormonal control with the feedback switch.

Self-test prompts

- What triggers ovulation?
- What does the corpus luteum produce?
- How do combined oral contraceptives prevent pregnancy?

Why This Matters to Your Patients

Polycystic ovary syndrome (PCOS): irregular cycles, anovulation, hyperandrogenism, insulin resistance; common cause of infertility. Endometriosis: endometrial tissue outside the uterus producing painful, cyclic bleeding and infertility. Menopause: ovarian estrogen production declines; hot flashes, bone loss, mood changes. Hormonal contraception, fertility treatment, and management of menstrual disorders all depend on understanding the axis. Pap smears, mammograms, and pelvic exams are screening tools nurses help administer. Pregnancy testing detects hCG.

TOPIC 3 OF 3 . WEEK 20 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Pregnancy and Development

Fertilization through the trimesters, plus the role of the placenta.

The Science

Fertilization happens in the ampulla of the uterine tube, usually within hours of ovulation. The zygote begins cleavage as it travels to the uterus, becomes a morula, then a blastocyst with an inner cell mass (becomes the embryo) and trophoblast (becomes the placenta). Implantation in the uterine endometrium is around day 6-7. Trophoblast secretes hCG, which maintains the corpus luteum so progesterone keeps the endometrium intact.

Embryonic period (weeks 1-8): all major organ systems form (organogenesis). Three germ layers (ectoderm, mesoderm, endoderm) differentiate and give rise to all tissues. Ectoderm: nervous system, epidermis. Mesoderm: muscle, bone, blood, kidney. Endoderm: GI lining, respiratory lining, glands. This is the period of highest vulnerability to teratogens because critical structures are forming.

Fetal period (week 9 to birth): growth and maturation. The placenta is the interface between maternal and fetal circulations; gases, nutrients, and wastes diffuse across without direct mixing of blood. The placenta is also a major endocrine organ in later pregnancy, producing estrogen, progesterone, hCG, and human placental lactogen.

Trimesters: first (weeks 1-12, organogenesis, high teratogen risk), second (weeks 13-26, rapid growth, fetal movement felt), third (weeks 27 to birth, lung and CNS maturation, weight gain). Surfactant production by Type II pneumocytes ramps up in the final weeks; premature birth before 34 weeks risks respiratory distress syndrome.

Teaching This Topic

Before the video (drawing prompt)

Have students list the three germ layers and one major tissue from each. Then outline the three trimesters with one developmental highlight per trimester.

Common misconceptions to address

- Students think fertilization happens in the uterus. It usually happens in the uterine tube (ampulla).
- The placenta is sometimes thought to mix maternal and fetal blood. It does not; the two circulations are separated by the placental membrane.
- All teratogenic damage is equal. The same teratogen has very different effects depending on when in development it is encountered.

Order of operations (what to teach first)

Fertilization, cleavage, implantation, then organogenesis with the germ layers, then placental function and the three trimesters.

Self-test prompts

- Where does fertilization typically occur?
- Name the three primary germ layers and one structure from each.
- Why is the first trimester the most vulnerable to teratogens?

Why This Matters to Your Patients

Ectopic pregnancy (implantation outside the uterus, usually in the uterine tube) is a surgical emergency; rupture causes hemorrhagic shock. Spontaneous abortion is common in the first trimester, often from chromosomal abnormalities. Pre-eclampsia is a third-trimester syndrome of hypertension and proteinuria, often progressing to eclampsia (seizures) if untreated. Gestational diabetes increases fetal size and delivery complications. Neonatal respiratory distress syndrome in preterm infants is treated with exogenous surfactant. Folate supplementation in early pregnancy reduces neural tube defects. Obstetric and neonatal nursing is built on this physiology.